Fertilization

Fertilization of the egg usually takes place in the fallopian tube within a day or so of ovulation. Although many sperm may be present, the penetration of a single sperm into the ovum sets up a series of biochemical events that usually prevent the entry of other sperm.

Fertilization is followed by the completion of meiosis II, with the formation of the second polar body (see Fig. 2-17). The chromosomes of the now-fertilized egg and sperm form **pronuclei,** each surrounded by its own nuclear membrane. It is only upon replication of the parental genomes after fertilization that the two haploid genomes become one diploid genome within a shared nucleus. The diploid **zygote** divides by mitosis to form two diploid daughter cells, the first in the series of cell divisions that initiate the process of embryonic development (see Chapter 14).

Although development begins at the time of conception, with the formation of the zygote, in clinical medicine the stage and duration of pregnancy are usually measured as the "menstrual age," dating from the beginning of the mother's last menstrual period, typically approximately 14 days before conception.

MEDICAL RELEVANCE OF MITOSIS AND MEIOSIS

The biological significance of mitosis and meiosis lies in ensuring the constancy of chromosome number—and thus the integrity of the genome—from one cell to its progeny and from one generation to the next. The medical relevance of these processes lies in errors of one or the other mechanism of cell division, leading to the formation of an individual or of a cell lineage with an abnormal number of chromosomes and thus an abnormal dosage of genomic material.

As we see in detail in Chapter 5, meiotic nondisjunction, particularly in oogenesis, is the most common mutational mechanism in our species, responsible for chromosomally abnormal fetuses in at least several percent of all recognized pregnancies. Among pregnancies that survive to term, chromosome abnormalities are a leading cause of developmental defects, failure to thrive in the newborn period, and intellectual disability.

Mitotic nondisjunction in somatic cells also contributes to genetic disease. Nondisjunction soon after fertilization, either in the developing embryo or in extraembryonic tissues like the placenta, leads to chromosomal mosaicism that can underlie some medical conditions, such as a proportion of patients with Down syndrome. Further, abnormal chromosome segregation in rapidly dividing tissues, such as in cells of the colon, is frequently a step in the development of chromosomally abnormal tumors, and thus evaluation of chromosome and genome balance is an important diagnostic and prognostic test in many cancers.

GENERAL REFERENCES

Green ED, Guyer MS, National Human Genome Research Institute: Charting a course for genomic medicine from base pairs to bedside, *Nature* 470:204–213, 2011.

Lander ES: Initial impact of the sequencing of the human genome, *Nature* 470:187–197, 2011.

Moore KL, Presaud TVN, Torchia MG: *The developing human: clinically oriented embryology*, ed 9, Philadelphia, 2013, WB Saunders.

REFERENCES FOR SPECIFIC TOPICS

Deininger P: Alu elements: know the SINES, *Genome Biol* 12:236, 2011.

Frazer KA: Decoding the human genome, *Genome Res* 22:1599–1601, 2012.

International Human Genome Sequencing Consortium: Initial sequencing and analysis of the human genome, *Nature* 409:860–921, 2001.

International Human Genome Sequencing Consortium: Finishing the euchromatic sequence of the human genome, *Nature* 431:931–945, 2004.

Venter J, Adams M, Myers E, et al: The sequence of the human genome, *Science* 291:1304–1351, 2001.

PROBLEMS

1. At a certain locus, a person has two alleles, *A* and *a*.
   a. What alleles will be present in this person's gametes?
   b. When do *A* and *a* segregate (1) if there is no crossing over between the locus and the centromere of the chromosome? (2) if there is a single crossover between the locus and the centromere?

2. What is the main cause of numerical chromosome abnormalities in humans?

3. Disregarding crossing over, which increases the amount of genetic variability, estimate the probability that all your chromosomes have come to you from your father's mother and your mother's mother. Would you be male or female?

4. A chromosome entering meiosis is composed of two sister chromatids, each of which is a single DNA molecule.
   a. In our species, at the end of meiosis I, how many chromosomes are there per cell? How many chromatids?
   b. At the end of meiosis II, how many chromosomes are there per cell? How many chromatids?
   c. When is the diploid chromosome number restored? When is the two-chromatid structure of a typical metaphase chromosome restored?

5. From Figure 2-7, estimate the number of genes per million base pairs on chromosomes 1, 13, 18, 19, 21, and 22. Would a chromosome abnormality of equal size on chromosome 18 or 19 be expected to have greater clinical impact? On chromosome 21 or 22?